

Revealing and Defending Against Backdoor Threats in Short-Term Federated Learning

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Abstract—Existing works on federated learning (FL) backdoor attacks often assume FL as a long-term process, where the global model is continuously trained on a dynamic dataset, and attackers launch rapid attacks near the convergence stage of the global model. However, in practical scenarios, FL can also be a short-term process. When static datasets are used, the training process may be terminated once the global model approaches convergence. For this scenario, we propose Early Stage Backdoor Attack (ESBA), a stealthy federated learning backdoor attack designed for the phase before global model convergence. The success of ESBA lies in utilizing a custom loss function that mitigates local forgetting, backdoor model parameter update constraints based on global model training inertia, and a centralized optimization mechanism for backdoor model updates that leverages cosine similarity and L2 norm. In addition, we observe that FL backdoor attacks exhibit abnormal statistical patterns in specific parameter dimensions. Motivated by this insight, we propose a novel anomaly detection metric and design Parameter Distribution Filtering Mechanism (PDFM). PDFM leverages HDBSCAN clustering on historical model updates from clients to identify malicious updates with statistically anomalous characteristics. We conduct comprehensive evaluations of ESBA and PDFM on six datasets. Experimental results show that ESBA can bypass nine advanced defense mechanisms. Moreover, PDFM can effectively defend against six advanced FL backdoor attack methods, including ESBA.

Index Terms—Federated learning, backdoor attack, robust federated learning.

I. INTRODUCTION

FEDERATED Learning (FL) was first introduced by McMah et al. [1] to enable multiple clients to collaboratively

solve machine learning problems under the coordination of a central server. This technology is gradually reshaping our daily lives, with FL-based applications emerging in various fields such as healthcare [2], [3], smart cities [4], [5], and the Internet of Things (IoT) [6], [7]. Unfortunately, FL has been shown to be vulnerable to backdoor attacks [8]. In such attacks, attackers upload malicious model updates containing the backdoor, known as backdoor model updates, to the server. These malicious updates implant the backdoor into the global model through aggregation. The backdoor causes the global model to misclassify samples containing specific triggers into the attacker's desired target class while not affecting its performance on datasets without the triggers.

Prior FL backdoor attack works often focus on long-term FL scenarios, where the training process can continue with dynamic data even after the global model approaches convergence. However, studies on practical FL systems show that computation and communication overheads remain major bottlenecks in edge and IoT scenarios, and early stopping can effectively reduce redundant resource consumption with limited accuracy sacrifice [9], [10]. For example, as the lack of widespread computational capability remains a barrier to the mass adoption of FL in real-world healthcare settings, most existing works in this domain adopt fixed-round training and may terminate training early once the global model reaches the expected performance criterion [11]. This paper studies backdoor threats and defenses in short-term FL, where the training process is terminated once the global model reaches a predefined convergence or fixed-round. The short-term scenario requires the attacker to complete the backdoor injection before the global model approaches convergence, whereas the works for the long-term process scenario often focus on launching attacks when the global model is near convergence to implant a persistent backdoor. The new challenges are that benign model updates are no longer close to zero [12] and the boundary of the difference interval between benign and backdoor model updates is larger [13].

Based on the above considerations, this paper proposes a novel stealthy FL backdoor attack method named Early Stage Backdoor Attack (ESBA), which can persistently attack the entire FL training process without affecting the benign performance of the model and can bypass defense mechanisms to complete the backdoor injection before the global model approaches convergence. The key design of ESBA is to guide

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backdoor update generation with the global model's training inertia, rather than merely constraining the final malicious updates by predefined stealth metrics. Training inertia denotes the accumulated short-term update direction of the global model, which characterizes the tendency of certain parameter dimensions to evolve consistently across recent FL training rounds. ESBA first trains the backdoor model using the custom constraint loss and standard classification loss. Then, it constrains the backdoor model updates based on the training inertia of the global model. Finally, ESBA performs centralized optimization on the backdoor model updates obtained from all malicious clients.

In the field of defending against FL backdoor attacks, anomaly detection and filtering have emerged as one of the core defense strategies [14]. Recent stealthy FL backdoor attacks have successfully bypassed traditional anomaly detection metrics by employing techniques such as loss constraints, highlighting the need to explore new detection indicators [15]. The essence of anomaly detection lies in distinguishing backdoor model updates from benign ones. This paper finds that the statistical distribution of specific parameters in the historical backdoor updates from malicious clients differs significantly from those of benign clients. Specifically, we first aggregate the historical model updates of clients, then extract the parameters in dimensions that deviate from the global model's training inertia direction, and finally use the mean and standard deviation of the fitted distribution of these parameters as anomaly detection indicators. Building on this, we propose a novel FL backdoor defense mechanism, the Parameter Distribution Filtering Mechanism (PDFM), which utilizes the proposed anomaly indicators in conjunction with the HDBSCAN clustering algorithm [16] to identify backdoor model updates.

Our contributions are summarized as follows:

- We investigate the problem of backdoor attacks in short-term FL scenarios and propose a novel FL backdoor attack method, ESBA. ESBA can persistently launch attacks throughout the entire FL training process and successfully inject backdoors before the global model approaches convergence.
- We explore a new parameter distribution-based anomaly detection metric for identifying backdoor model updates and further propose a novel FL defense mechanism, PDFM, by combining this metric with the HDBSCAN clustering algorithm.
- We conduct extensive experiments on six benchmark datasets, using five advanced FL backdoor attack methods and nine advanced defense mechanisms to comprehensively evaluate the effectiveness of our proposed attack and defense methods. Experimental results show that ESBA can effectively perform short-term FL backdoor attacks, and PDFM effectively defends against six advanced backdoor attack methods, including ESBA.

The remainder of this paper is organized as follows: Section II discusses related works about this paper. Section III introduces the necessary background knowledge and the threat model. Section IV elaborates on the proposed attack method,

ESBA. Section V elaborates on the proposed defense mechanism, PDFM. Section VI presents and analyzes the experimental results. Finally, Section VII concludes this paper.

II. RELATED WORK

A. Backdoor Attacks in FL

The initial research on backdoor attacks in FL aims to expose the backdoor attack threats faced by the FL framework. Bagdasaryan et al. [12] theoretically analyzed that backdoor attacks are more likely to succeed when the global model in FL is close to convergence and proposed a backdoor attack method based on model gradient replacement. Wang et al.'s work [17] provided a theoretical analysis demonstrating the susceptibility of FL to backdoor attacks and proposed a method that utilizes the tails of input data distributions for backdoor attack. Xie et al. [18] elucidated the threat of distributed backdoor attacks in FL. Researchers have also investigated backdoor attack methods in special FL scenarios, including personalized FL [19], vertical FL [20], [21], and graph FL [22]. Subsequent works focused on designing new mechanisms to enhance the persistence or stealth of FL backdoor attack methods. Persistence refers to the ability of the implanted backdoor in the global model to survive as many training rounds as possible. Zhang et al. [23] proposed that modifying only a small subset of parameters with the lowest values in the global model during backdoor model training can ensure good persistence of the implanted backdoor. Gong et al. [24] pointed out that subsequent benign updates quickly weaken the effect of a single backdoor trigger. Dai et al. [25] found that benign images with target labels of both original and poisoned images have a crucial impact on the persistence of backdoors. Liu et al. [26] designed a novel attack persistence metric and the Fully-Combination Backdoor Attack (FCBA) method with good persistence.

On the other hand, work focused on stealth aims to maximize the attack's ability to confront FL backdoor defense mechanisms. Li et al. [15] addressed deep defenses against detecting backdoor models from different perspectives by optimizing backdoor models for multiple targets and employing noise masking and induction model mechanisms. Lyu et al. [13] established lower and upper bounds on local model changes caused by backdoor triggers for malicious participants in general FL tasks, ensuring the stealth of attack methods by controlling critical factors of local model biases induced by backdoor triggers. Zhang et al. [27] and Nguyen et al. [28] customized dynamic triggers based on the training process of the global model, successfully bypassing various defense mechanisms and achieving good persistence. However, the time complexity of dynamic trigger generation methods is significantly higher than that of backdoor model update optimization methods. Moreover, existing dynamic trigger generation techniques are limited to supporting image recognition tasks.

Our work focuses on stealthy FL backdoor attacks and builds the attack method based on static triggers. We draw on the mainstream idea of FL backdoor attacks using static triggers that introduce custom constraint losses in the backdoor

model and add additional perturbations to the backdoor model updates.

B. Backdoor Defenses in FL

There are two mainstream ideas to designing defense mechanisms against FL backdoor attacks. One idea is distinguishing backdoor model updates through anomaly detection based on specific similarity metrics. Widely used similarity metrics include euclidean distance, cosine similarity, and update direction. Fung et al. [29] observed high pairwise cosine similarity between backdoor model updates and designed Foolsgold based on this observation. Ozdayi et al. [30] used the difference in update directions between backdoor model updates and benign model updates. Zhang et al. [31] filtered anomalous model updates by examining the consistency of historical model updates. Huang et al. [32] identified backdoor model updates by considering three similarity metrics: Manhattan distance, Euclidean distance, and cosine similarity. Rieger et al. [33] also employed three different distance matrices to measure the distance between each pair of model updates and clustered model updates based on these matrices.

Another mainstream idea is to design robust aggregation algorithms. Sun et al. [34] proposed norm clipping and differential privacy mechanisms to mitigate backdoor attacks without affecting global model training. Rodríguez-Barroso et al. [35] calculated each dimension's mean and standard deviation and pruned all model updates based on these two metrics. Nguyen et al. [36] combined above two ideas to design a sophisticated defense mechanism, FLAME. FLAME filters malicious model updates based on cosine similarity detection metrics and uses adaptive clipping and noise mechanisms for robust aggregation. Huang et al. [37] proposed Lockdown, an isolated subspace training method that mitigates the poison-coupling effect in FL by isolating client training subspaces and applying quorum consensus. Miao et al. [38] proposed an efficient and secure FL scheme based on adaptive Local Differential Privacy and compressed sensing to defend against backdoor attacks.

We observed significant differences in the parameter distribution between backdoor model updates and benign model updates in specific dimensions. Based on this difference and aggregation strategies, we designed the anomaly detection mechanism PDFM to filter out backdoor model updates. The design of PDFM draws on the ideas from existing works that consider historical model updates [29], [31] and employ clustering algorithms [33], [36] to identify anomalous model updates.

III. PRELIMINARY

A. Problem Formulation

Federated Learning: This paper adopts a typical FL task. Specifically, in training round t , client i first receives the global model G_{t-1} obtained from the previous round of aggregation algorithm from the server. Then, client i trains on its private data $\mathcal{D}_i$ to obtain a local model $W_{t,i}$. The training process can be formalized as follows:

$$W_{t,i} = G_{t-1} - lr_i \cdot \nabla \mathcal{L}(G_{t-1}, \mathcal{D}_i) \quad (1)$$

where lr_i represents the local learning rate of client i , and $\nabla \mathcal{L}(G_{t-1}, \mathcal{D}_i)$ denotes the cross-entropy loss gradient of model G_{t-1} and data $\mathcal{D}_i$. Afterward, each client uploads the trained local model $W_{t,i}$ or local model update $\Delta W_{t,i} = W_{t,i} - G_{t-1}$ to the server to participate in the new round of the aggregation process. Taking the classical mean aggregation algorithm with a global learning rate of η as an example, the process above can be formalized as:

$$G_t = G_{t-1} + \frac{\eta}{N} \cdot \sum_{i=1}^N (W_{t,i} - G_{t-1}) \quad (2)$$

Backdoor Attack: In any given round t , the attacker begins by selecting a portion of samples from their dataset and embedding triggers into them. These poisoned samples are combined with the remaining clean data in the local dataset to form the final local poisoned dataset $\mathcal{D}^P = \{\mathcal{D}_{clean}, \mathcal{D}_{poison}\}$. The attacker then uses $\mathcal{D}^P$ to train the local backdoor model $W_{t,adv}$. The training process is as follows:

$$W_{t,adv} = G_{t-1} - lr_i \cdot \nabla (\mathcal{L}(G_{t-1}, \mathcal{D}^P) + \beta \cdot \mathcal{L}_{cons}) \quad (3)$$

where $\mathcal{L}_{cons}$ refers to the constraint loss imposed by the attacker on the backdoor model, the latest backdoor attack methods typically customize $\mathcal{L}_{cons}$ for the attack goal to enhance attack performance. After training the backdoor model, the attacker uploads $W_{t,adv}$ or $\Delta W_{t,adv} = W_{t,adv} - G_{t-1}$ to the server to participate in aggregation, thereby manipulating the parameters of the global model and implanting the backdoor.

Backdoor Defense: Defense mechanisms against FL backdoor attacks primarily include two strategies: anomaly detection and robust aggregation. A typical defense mechanism can employ either any single strategy or a combination of both. Anomaly detection methods assign different weight γ_i ($\gamma_i \in [0, 1]$) to model update W_i , with update exhibiting greater anomalies receiving lower weight. Robust aggregation processes each model update according to specific aggregation strategies. They can be formally described as follows:

$$G_t = G_{t-1} + Robust_aggr(\gamma \cdot \mathbb{W}_t) \quad (4)$$

Here, $Robust_aggr()$ represents the robust aggregation strategy, $\mathbb{W}_t = \{\Delta W_{t,1}, \Delta W_{t,2}, \dots, \Delta W_{t,N}\}$ denotes the set of all model updates uploaded by clients, and γ refers to the weight vector composed of γ_i corresponding to each model update.

B. Threat Model

Attack Goal: The attacker's goal is for the attacked global model to misclassify poisoned samples containing triggers into the target label specified by the attacker, while clean samples without triggers are correctly classified. To achieve this goal, the attacker must counter various backdoor defense mechanisms that the server may have deployed. Additionally, the attacker needs to achieve the desired attack effect before the global model approaches convergence.

Attack Capability: This paper considers the same limited-knowledge FL backdoor attack scenario as the work of Li et al. [15]. Specifically, this paper assumes that the attacker can only control a small number of the N clients in the FL system,

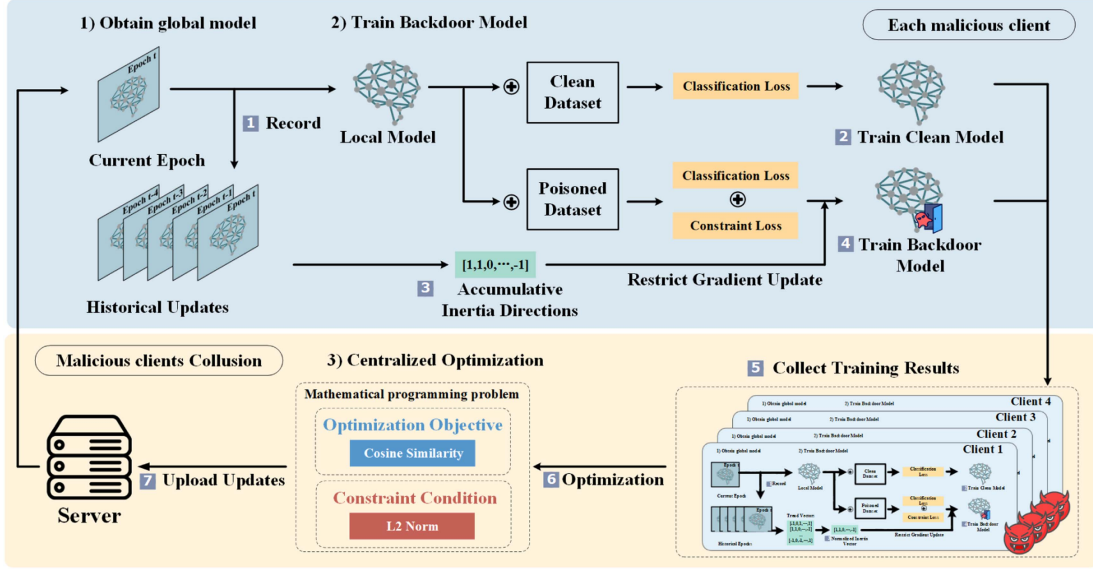


Fig. 1. The overall workflow of ESBA.

turning them into malicious clients for conducting attacks. The attacker has complete control over these malicious clients, including access to and modification of local datasets, the ability to customize local model training rules, and access to the global model. The attacker also can coordinate these malicious clients to collaborate with each other. However, the attacker cannot access benign clients to obtain their local data or information about their local model training processes. While the attacker can upload carefully crafted backdoor model updates to the server for aggregation, they are unaware of the defense mechanisms deployed on the server and cannot access the model updates uploaded by the benign clients.

IV. ATTACK METHOD

A. Overview

Fig. 1 shows the workflow of ESBA and Algorithm 1 outlines the procedure of ESBA. ESBA consists of three coupled components. First, each malicious client trains a local backdoor model with a constraint loss that discourages large modifications to parameters important for the original task. Second, ESBA applies an inertia poisoning mechanism, which retains only gradient dimensions whose directions are consistent with the accumulated recent global update direction. This makes the malicious update follow the short-term optimization trend of the global model. Third, malicious clients collaboratively perform centralized optimization on their backdoor updates. This step maximizes the similarity between backdoor updates and clean local updates while constraining the aggregate perturbation by an L2-norm condition. These three components jointly enable ESBA to inject a backdoor before global convergence while reducing the chance of being detected by existing defenses.

B. Backdoor Model Loss Function

Considering any client i performing the task of minimizing the local loss as described in (1), this minimization of local loss is equivalent to maximizing the corresponding posterior probability [39]. It can be formalized as $\min_{\theta} \mathcal{L}(\theta) = \max_{\theta} \mathcal{P}(\theta|\mathcal{D}_i)$. Here, θ represents the local model parameters of the client. According to Bayesian theory, the posterior probability of client i can be decomposed into a likelihood $\ln \mathcal{P}(\mathcal{D}_i|\theta)$, a prior $\mathcal{P}(\theta)$, and a constant $\mathcal{P}(\mathcal{D}_i)$ when the dataset $\mathcal{D}_i$ is given, as shown below:

$$\ln \mathcal{P}(\theta|\mathcal{D}_i) = \ln \mathcal{P}(\theta) + \ln \mathcal{P}(\mathcal{D}_i|\theta) - \ln \mathcal{P}(\mathcal{D}_i) \quad (5)$$

Furthermore, in continuous learning, the posteriors of old tasks can also be treated as priors when learning a new task. So, the posterior of the globally aggregated model from the previous round can be used as the prior for the current round's client training $\mathcal{P}(\theta)$. Since the global model is formed by aggregating local models, the global posterior can be estimated using the local posteriors. In practice, a Gaussian distribution can be used to approximate the posteriors in a small neighborhood of the MAP estimate on single clients [40], [41]. Assuming the global posterior distribution follows a Gaussian distribution, denoted as $\mathcal{P}_G(\theta|\mathcal{D}) \approx \mathcal{N}(\theta|\mu_G, \Sigma_G)$, where μ_G and Σ_G represent the mean and covariance, respectively. Then the posterior probability of client i can be expressed as:

$$\ln \mathcal{P}(\theta|\mathcal{D}_i) = \ln \mathcal{P}_G(\theta|\mathcal{D}) + \ln \mathcal{P}(\mathcal{D}_i|\theta) - \ln \mathcal{P}(\mathcal{D}_i) \quad (6)$$

Existing work [39] has shown that constraining the prior loss can mitigate local forgetting, and according to the second-order regularization term derived directly from $\mathcal{P}_G(\theta|\mathcal{D})$, the prior loss can be formulated as $\mathcal{L}_{prior} = -\ln \mathcal{P}_G(\theta|\mathcal{D}) = \frac{1}{2}(\theta - \mu_G)^T \Sigma_G^{-1}(\theta - \mu_G)$. Refer to the works of Kirkpatrick et al. [42] and Shoham et al. [43], Σ_G^{-1} reflects the importance of each dimension in the global model, and by minimizing

$\frac{1}{2}(\theta - \mu_G)^T \sum_G^{-1}(\theta - \mu_G)$, the modifications to important dimensions in the client model can be effectively suppressed, limiting the distance between the local model and the global model. This enables clients to learn to solve their sub-problems without significantly affecting the global task performance of the model. This constraint idea is equally applicable to our backdoor attack problems. Some works [39], [42], [43] have proposed methods for the server to estimate μ_G and $\sum_G^{-1}$. These specific methods are only applicable to model trainers who can grasp the complete information of the training process, and attackers in black-box scenarios cannot use them. Therefore, we explore a new similar constraint loss to achieve the idea of the above prior loss, which is formally described as follows:

$$\mathcal{L}_{cons} = \sum_{k=1}^K \frac{1}{2} \cdot \nabla \mathcal{L}_k(W_{e-1,k}, \mathcal{D}_i^p) \cdot (W_{e-1,k} - G_{t-1,k})^2 \quad (7)$$

where e represents the local training round for the client, t represents the server's global model training round, and K represents the number of model parameters. $\nabla \mathcal{L}_k(W_{e-1,k}, \mathcal{D}_i^p)$ refers to the cross-entropy loss of the backdoor model on the poisoned dataset. ESBA uses the global model parameters from the previous round of aggregation to approximate μ_G . It also uses the gradient of the backdoor model's loss function on poisoned data with respect to the model parameters to reflect the importance of each dimension, achieving a similar effect to $\sum_G^{-1}$. This constraint loss suppresses the modifications to the global task's important dimensions, embedding the backdoor in positions of the global model that are less relevant to the global task. Zhang et al.'s work [23] revealed that benign clients have relatively small updates for unimportant parameters, making it easier for attackers to manipulate the direction and magnitude of these parameter updates. The constraint loss also suppresses the modifications to the important dimensions relevant to the backdoor task, preventing tremendous parameter value change in the dimensions where the backdoor is implanted. This reduces the difference between the backdoor model and the benign model, which helps evade anomaly detection in defense mechanisms. The complete loss function used by the attacker to train the backdoor model is as follows:

$$\mathcal{L}_{backdoor} = \mathcal{L}_{task} + \alpha \mathcal{L}_{cons} \quad (8)$$

where α is used to adjust the weight of the constraint loss, and $\mathcal{L}_{task}$ refers to the cross-entropy loss function of the model on the poisoned dataset.

C. Inertia Poisoning Mechanism

Existing work [44] has demonstrated that methods leveraging the inertia of model training to promote model optimization are also applicable during the training of global models in FL. Inspired by this, we propose leveraging the inertia of the global model training process to facilitate backdoor implantation and counter defense mechanisms. Specifically, ESBA tracks the direction trend of changes in the global model over the previous historical training stage S . When ESBA performs stochastic gradient descent optimization on the backdoor model $W_{t,adv}$,

it retains only the gradient updates that align with the direction of accumulative changes in the global model over the previous historical training stage. This process can be formalized as follows:

$$W_{t,adv} = G_{t-1} - lr_b \nabla \mathcal{L}_{backdoor} \cdot eq(sign(vector_{acc}), sign(\nabla \mathcal{L}_{backdoor})) \quad (9)$$

where t represents the backdoor model training round, lr_b and $\nabla \mathcal{L}_{backdoor}$ denote the backdoor model learning rate and loss gradient, respectively. $sign()$ is used to compute the sign of each element in the input. $eq()$ is then used to compare whether the elements of two inputs are equal in each dimension. $vector_{acc}$ represents how many rounds of historical gradient changes are accumulated to represent inertia. $vector_{acc}$ represents the accumulated update direction vector from the previous historical stage, and the cumulative result of the first stage can be expressed as:

$$vector_{acc} = \sum_{s=0}^{S-1} sign(G_{s+1} - G_s) \quad (10)$$

In a black-box scenario, the attacker cannot access the model updates of other benign clients. However, leveraging the inertia characteristic of global model training, the attacker can reduce the difference between the backdoor model updates and benign model updates by aligning the backdoor model updates with the inertia of the global model updates.

D. Centralized Optimization of Backdoor Model Updates

At this stage, ESBA aims to mitigate the potential adverse effects it may have during the backdoor model training process and further enhance its ability to counteract defense mechanisms. When training the backdoor model, ESBA uses constraint loss, which results in more minor modifications in dimensions that are more important for the original task. However, these dimensions are the focus of other benign updates, potentially creating a significant distinction between backdoor and benign model updates. ESBA establishes the process of centralized optimization of backdoor model updates as a constrained minimization problem. Specifically, the constraint conditions and optimization objectives of an arbitrary backdoor model update Δw_i can be formally expressed as follows:

$$\begin{cases} \min & (1 - cosine_similarity(\Delta w_i, W_{i,c})) \\ \text{s.t.} & L2_norm\left(\sum_{j=1}^M \Delta w_j - \sum_{j=1}^M \Delta w_{j,0}\right) = 0 \end{cases} \quad (11)$$

where M denotes the number of malicious clients, $W_{i,c}$ represents the benign model trained by the i -th malicious client using clean samples without triggers, Δw_j indicates the backdoor model update by the j -th malicious client, $\Delta w_{j,0}$ denotes the backdoor model update before optimization.

The reason for using the cosine similarity metric is that it can mitigate the adverse effects induced by the constraint loss. Specifically, the formula for calculating cosine similarity can be expressed as $cosine_similarity(\Delta w_i, W_{i,c}) = \frac{\Delta w_i \cdot W_{i,c}}{\|\Delta w_i\| \cdot \|W_{i,c}\|}$.

Since $W_{i,c}$ and $\|W_{i,c}\|$ are constants, increasing cosine similarity is equivalent to increasing $\frac{\Delta w_i \cdot W_{i,c}}{\|\Delta w_i\|}$. The key to increasing $\frac{\Delta w_i \cdot W_{i,c}}{\|\Delta w_i\|}$ lies in achieving the highest possible value of $\Delta w_i \cdot W_{i,c}$ while keeping $\|\Delta w_i\|$ constant. To achieve this goal, the parameter values of Δw_i in the dimensions where $W_{i,c}$ has larger values should also be larger. These dimensions, where $W_{i,c}$ has higher parameter values, are more critical for the original task. By increasing the values of Δw_i in these dimensions, it compensates for the issue where the constraint loss might cause smaller parameter values in these dimensions, which could otherwise lead to a noticeable distinction from benign model updates. Moreover, to maintain the ratio between $\Delta w_i \cdot W_{i,c}$ and $\|\Delta w_i\|$, the optimization mechanism reduces the values of Δw_i in the dimensions where $W_{i,c}$ has smaller parameter values. This further minimizes the difference between Δw_i and other benign model updates.

The purpose of adopting centralized optimization rather than optimizing individually for each client is to apply a global constraint on the variation of backdoor model updates using the L2 norm, such that the perturbations introduced to each backdoor model update at this stage can cancel each other out through the aggregation mechanism. Even if the defense mechanism assigns different weights to the model updates, preventing the noise from fully canceling out after aggregation, the L2 constraint ensures that the final noise applied to the global model is small. The work of Sun et al. [34] shows that adding slight noise does not affect the effectiveness of FL. The term noise refers to the optimization vector applied to the backdoor model updates by the centralized optimization mechanism based on (11). Although the optimization vector helps enhance the stealthiness of the backdoor updates, it is considered noise with respect to the backdoor task.

V. DEFENSE MECHANISM

Due to the distinct differences between backdoor and original tasks, the critical dimensions of backdoor model updates and benign model updates vary. Previous studies [23] have pointed out that parameter values in model updates tend to be larger in dimensions that are more essential for the respective task. Based on this observation, if certain dimensions can be identified as critical to benign model updates but not to backdoor model updates, then examining the parameter distribution in these dimensions can effectively distinguish backdoor updates from benign updates. The inertia characteristics of the global model discussed in Section IV-C offer a practical approach to locating these dimensions. Since attackers typically control only a limited number of FL clients to perform backdoor attacks [45], the global model's training inertia reflects essential optimization directions for the original task. Dimensions that have the same directions as this inertia indicate are more significant for benign models, though potentially less significant for backdoor models with different objectives. PDFM thus uses the statistical distribution of parameters in these dimensions as the key distinguishing feature.

Algorithm 1: ESBA.

Input: Weight of the constraint loss α , Malicious Clients number M , Local Dataset $\mathcal{D}_i$, Learning rate lr_b , Local model training rounds E , Centralized Optimization rounds R

Output: Backdoor model updates Δw

```

1: for  $i = 1, 2, \dots, M$  do
2:   Obtain the global model  $G_{t-1}$ 
3:   Train benign model  $W_{i,c}$  using clean datasets  $\mathcal{D}_i$ ;
4:   for  $e = 1, 2, \dots, E$  do
5:     // Train the backdoor model  $W_i$  based on (7), and (8);
6:      $\mathcal{L}_{backdoor} = \mathcal{L}_{task}(W_{e-1,i}, \mathcal{D}_i^P) + \alpha \sum_{k=1}^K \frac{1}{2} \cdot \nabla \mathcal{L}_k(G_{t-1}, \mathcal{D}_i^P) \cdot (W_{e-1,k,i} - G_{t-1,k})^2$ 
7:      $g_{e,i} = \nabla \mathcal{L}_{backdoor}$ 
8:     // Project gradient onto inertia constraint based on (9), and (10);
9:      $W_{e,i} = W_{e-1,i} - lr_b \cdot eq(sign(vector_{acc}), sign(g_{e,i})) \cdot g_{e,i}$ 
10:   end for
11:    $\Delta w_i = W_i - G_{t-1}$ 
12: end for
13: for  $r = 1, 2, \dots, R$  do
14:   // Optimize all backdoor model updates centrally using (11);
15:   for  $i = 1, 2, \dots, M$  do
16:      $\mathcal{L}_{norm} = L2\_norm(\sum_{j=1}^M \Delta w_j - \sum_{j=1}^M \Delta w_{j,0})$ 
17:      $\mathcal{L}_{cos} = 1 - cosine\_similarity(\Delta w_i, W_{i,c})$ 
18:      $\mathcal{L} = \mathcal{L}_{norm} + \mathcal{L}_{cos}$ 
19:      $\Delta w_{r,i} = \Delta w_{r,i} - lr_b \cdot \nabla \mathcal{L}$ 
20:   end for
21: end for
22: Upload backdoor model updates  $\Delta w_i (i = 1, 2, \dots, M)$  to the server;

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Fig. 2 shows the workflow of PDFM and Algorithm 2 outlines the procedure of PDFM. PDFM leverages the cumulative inertia direction vector $vector_{acc}$, as described in (10), to identify dimensions in model updates that are the same as the global model's inertia direction. To further amplify statistical distribution differences in these dimensions, PDFM maintains a history of each client's uploaded model updates, allowing the aggregation of historical updates to increase parameter value gaps. Additionally, existing work [29] demonstrates that directly clustering historical aggregated gradients is insufficient to distinguish between benign and backdoor model updates. Therefore, similarity metrics are necessary to further extract the differences between them. Rodriguez et al. [35] demonstrate that statistical indicators, such as mean and standard deviation, can be used to capture the statistical differences between benign and backdoor model updates. Therefore, PDFM characterizes statistical distributions with the mean and variance metrics and uses these two statistics to filter backdoor model updates. Similar to advanced defenses like FLAME and DeepSight, PDFM employs the HDBSCAN algorithm for anomaly detection. HDBSCAN clusters

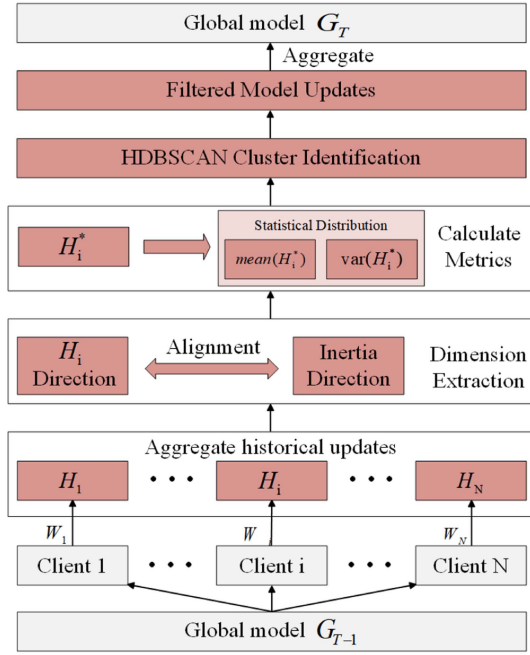


Fig. 2. Illustration of PDFM's workflow in epoch T.

Algorithm 2: PDFM.

Input: Clients number N , Local Model Updates $\Delta w_i (i = 1, 2, \dots, N)$, The number of training epoch T , Historical model updates from client i at epoch t $\Delta w_{i,t}$, Cumulative inertia direction $vector_{acc}$ based on (10)

Output: The global model after this epoch of aggregation G_T

```

1: for All client  $i$  do
2:   Let  $H_i$  be the aggregate historical vector  $\sum_{t=1}^T \Delta w_{i,t}$ 
3:    $indexs = where(eq(sign(\Delta w_i), sign(vector_{acc})))$ 
4:    $H_i^* = H_i[indexs]$ 
5:    $d_i = (mean(H_i^*), var(H_i^*))$ 
6: end for
7:  $clusters = HDBSCAN(d_1, d_2, \dots, d_N)$ 
8: for client  $j$  in  $cluster_{max}$  do
9:    $G_T = G_{T-1} + lr \cdot \Delta w_j / |cluster_{max}|$ 
10: end for

```

model updates based on the mean and variance of parameter distributions in specific dimensions and can dynamically determine the required number of clusters. The model updates classified as benign are those associated with the statistical distribution metrics within the largest cluster result obtained through the dynamic clustering algorithm HDBSCAN. The server ultimately aggregates these benign model updates to obtain the new global model.

Let N_t denote the number of participating clients in round t , N denote the total number of clients, and K denote the number of model parameters. For each round, PDFM incrementally maintains the historical update vector H_i , selects dimensions

according to the inertia direction, and computes the mean and variance of the selected dimensions. These operations require $O(N_t \cdot K)$ time. After feature extraction, each client is represented by only two statistical features and HDBSCAN is applied to these low-dimensional features. The clustering costs $O(N_t^2)$ time. Therefore, the per-round time complexity of PDFM is $O(N_t \cdot K + N_t^2)$. For space complexity, PDFM only stores one historical aggregated update vector for each client, resulting in $O(N \cdot K)$ memory. The auxiliary storage for clustering is at most $O(N_t^2)$. Therefore, the overall space complexity is $O(N \cdot K + N_t^2)$. These complexity results show that PDFM is scalable in large-scale FL deployments. In terms of time overhead, the parameter-level cost of PDFM is only $O(N_t \cdot K)$, which is comparable to the standard server-side aggregation cost because the server already needs to process N_t uploaded model updates in each round. Since practical large-scale FL systems usually adopt partial client participation, N_t is much smaller than the total number of clients N , making $O(N_t^2)$ manageable. Moreover, PDFM stores only one accumulated historical update vector for each client, resulting in $O(N \cdot K)$ memory without growing with the number of training rounds.

VI. EMPIRICAL EVALUATIONS

A. Experimental Setup

Dataset: We experimented with ESBA and PDFM using five benchmark datasets, namely CIFAR-10, CIFAR-100 [46], GT-SRB [47], Tiny-ImageNet [48], IMDB [49] and Obscene [50]. The CIFAR-10 dataset consists of 60,000 images in 10 classes, with 6,000 images per class. There are 50,000 training images and 10,000 test images. CIFAR-100 is an image classification dataset for computer vision with 100 categories, each containing 600 images (500 training images and 100 test images). GTSRB contains 43 classes of traffic signs, divided into 39,209 training images and 12,630 test images. The size of all the images in the above three datasets used in the experiments is 32×32 pixels. The Tiny-ImageNet dataset contains 200 classes, with 500 training images and 50 test images per class. Each image is resized to 64×64 pixels. The IMDB dataset contains 50,000 movie reviews, categorized into positive and negative classes, making it a commonly used text classification dataset. The Obscene dataset is sampled from the Civil Comments dataset. The Civil Comments dataset collects comments from approximately 50 English-language news sites worldwide and labels them for harmful content to facilitate malicious text detection tasks. We labeled comments from the Civil Comments dataset with an obscene value greater than 0.5 as containing obscene content and sampled an equal number of comments without obscene content to form the dataset used in this paper.

Attack Methods and Defense Mechanisms: We use four FL backdoor attack methods to compare with ESBA and evaluate the effectiveness of PDFM, namely Neurotoxin [23], 3DFed [15], CerP [13], FCBA [26], and ModelReplace [12]. ModelReplace represents the most classic backdoor attack method; 3DFed and CerP are the state-of-the-art stealthy backdoor attack methods based on static and dynamic triggers, respectively. Neurotoxin is a widely recognized advanced FL

backdoor attack method focusing on persistence. FCBA is a state-of-the-art distributed FL backdoor attack method. We also use eight state-of-the-art or representative defense mechanisms to compare with PDFM and evaluate the effectiveness of ESBA, namely Deepsight [33], FLAME [36], FLDetector [31], Fools-gold [29], MMetrics [32], RFOUT [35], RLR [30], SNOW-BALL [51], and NC+DP [34].

Triggers and Target Labels: For each image recognition dataset, the attacker uses the same static trigger as in the experiment by Li et al. [15]. The trigger is a 3×5 pixel pattern placed in the bottom left corner of the image. For the CerP method based on dynamic triggers, the configuration parameters of the dynamic trigger training process are the same as those in the experiment setup publicly disclosed by Lyu et al. [13]. The trigger patterns used in the CerP method and the FCBA method are the same as the experimental trigger patterns publicly released by Lyu et al. [13]. The target label set by the attacker for each dataset is 8. For the IMDB dataset, the trigger used by the attacker is the sentence, “This movie is highly favored by James Bond,” with the target label set to 0. The attacker aims to cause the target large language model to misclassify positive reviews with the trigger as negative. For the Obscene dataset, the attacker uses the same trigger as in the experiment by Wan et al. [52], replacing the main clause in the comments with “James Bond” and setting the target label to 0. The attacker’s objective is to cause the target large language model to misclassify obscene comments containing “James Bond” as not obscene.

Federated Learning Setup: For CIFAR-10, CIFAR-100, and GTSRB, we set up the FL framework like Li et al.’s experiment [15]. One hundred FL clients participated in the FL task, including ten malicious clients performing backdoor attacks. In each round of aggregation, the server selects all clients to participate in aggregation. We sample each client’s local dataset with a Dirichlet distribution to simulate non-IID data scenarios. Both local and global models use the ResNet-18 architecture. The learning rate of the global model is set to 1. Benign clients train local models for two rounds, while malicious clients train backdoor models for five rounds. Every local model and backdoor model is trained using the SGD optimizer with a learning rate set to 0.1. In the text classification tasks, there are twenty FL clients, four of which attackers can control to launch backdoor attacks. We sample each client’s local dataset evenly to simulate IID data scenarios. During each round of aggregation, all clients participate in the training. The server uses a BERT model as the global model for training, with a global learning rate set to 1. Both benign and malicious clients use the AdamW optimizer for training, with the learning rate and the number of local training epochs set to 0.00003 and 1, respectively. For Tiny-ImageNet, there are 100 clients in total, among which 10 are malicious. We sample each client’s local dataset with a Dirichlet distribution to simulate non-IID data scenarios. In each aggregation round, both benign and malicious clients have a 0.8 probability of successfully uploading their model updates to the server. The server randomly selects 10 updates from all successfully uploaded model updates for aggregation. Both local and global models use the ResNet-18 architecture. The learning rate for both benign and malicious clients is set to 0.001, while

the global model uses a learning rate of 1. For all tasks, the attack starts once ESBA completes its first accumulation of the inertia vector and continues until the end of FL training. The attack begins in the fifth round for image classification tasks and in the second round for text classification tasks. The default values of the centralized optimization round and the constraint loss weight are set to 10 and 1, respectively. For PDFM, the inertia direction vector accumulation rounds are set to 3 and 1 by default for image and text tasks, respectively.

Evaluation Metrics: We use two metrics to evaluate the effectiveness of attack and defense: backdoor task accuracy (BAC) and clean accuracy (CA). BAC assesses whether the backdoor attack successfully implants a backdoor in the target model or whether the defense mechanism successfully resists the backdoor attack. Specifically, BAC refers to the classification accuracy of the attacked global model on a poisoned test dataset with triggers for each image. CA is used to assess whether the backdoor attack will damage the performance of the target model on the original task. It refers to the classification accuracy of the attacked global model on a clean test dataset without triggers. For ESBA, the experiments aim to evaluate whether it interferes with the global model’s learning of the original task and its ability to bypass defense mechanisms. The former is assessed by observing whether continuously launching the ESBA attack throughout FL training significantly reduces the final global model’s CA compared to the no-attack scenario. The latter is evaluated by examining whether ESBA achieves a high BAC when facing various defense methods. For PDFM, the experiments aim to assess whether it disrupts global model training and its effectiveness in defending against backdoor attacks. The former is evaluated by checking whether deploying PDFM leads to a significant drop in the final global model’s CA compared to training without PDFM. The latter is assessed by observing whether the BAC remains low under backdoor attacks when PDFM is deployed.

B. Effect of ESBA

Figs. 3 and 4 illustrate the attack results on the FL framework without defense mechanisms using the CIFAR-10 and CIFAR-100 datasets, omitting BAC results for attacks that caused the target model to be unusable. Corresponding MAC results are shown in Table I. We emphasize MAC values in Table I that show substantial reductions due to attacks with arrows and use “failure” to denote cases where the global model’s usability drops to MAC below 20%. The experiments reveal that existing FL backdoor attack algorithms have limitations in attack timing, even when facing the FedAvg aggregation algorithm that lacks resistance to backdoor attacks. Specifically, when attacks are launched prematurely, they may fail to achieve the desired BAC for the implanted backdoor or significantly reduce the model’s MAC. As seen in Fig. 3, the backdoors implanted by Neurotoxin and CerP fail to reach the ideal BAC. Table I shows that ModelReplace and 3DFed notably degrade the model’s performance on the original task. The drop in CA for some baseline attacks is due to our experimental setting, which launches attacks throughout the entire training rounds of FL. In contrast, 3DFed

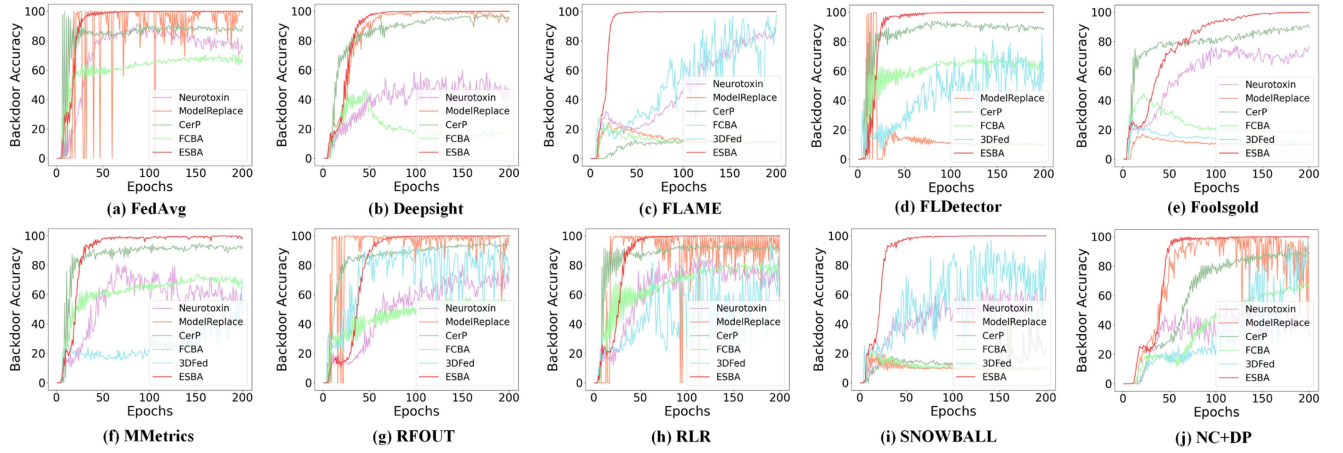


Fig. 3. BAC results of backdoor attacks that have acceptable CA result on CIFAR-10 (%).

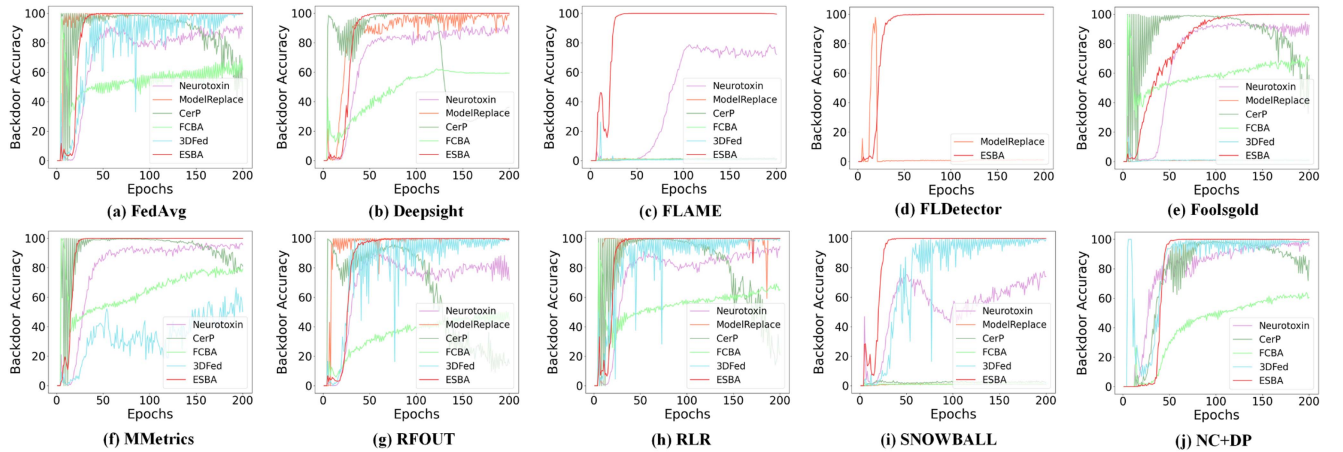


Fig. 4. BAC results of backdoor attacks that have acceptable CA result on CIFAR-100 (%).

TABLE I
CA RESULTS OF DIFFERENT BACKDOOR ATTACKS ON CIFAR-10 AND CIFAR-100. *failure* INDICATES THAT THE CA IS BELOW 20%, WHICH MEANS THE BACKDOOR ATTACK CAUSES THE GLOBAL MODEL TO BE UNACCEPTABLE.

	No Attack		Neurotoxin		ModelReplace		CerP		FCBA		3DFed		ESBA(Ours)	
	CIFAR10	CIFAR100												
FedAvg	81.1%	50.70%	80.74%	50.52%	76.65%↓	37.27%↓	79.97%	50.20%	80.40%	49.50%	<i>failure</i>	49.67%	83.45%	52.18%
Deepsight	80.25%	50.54%	79.15%	49.99%	78.53%	44.80%↓	67.28%↓	35.01%↓	67.20%↓	35.02%↓	<i>failure</i>	<i>failure</i>	83.23%	52.27%
FLAME	73.93%	41.58%	68.77%↓	42.01%	72.57%	40.98%	71.84%	41.01%	73.16%	41.88%	67.63%↓	41.33%	75.92%	42.01%
FLDetector	80.65%	50.82%	<i>failure</i>	<i>failure</i>	78.64%	37.41%↓	80.14%	<i>failure</i>	79.68%	<i>failure</i>	79.77%	<i>failure</i>	83.21%	52.13%
Foolsgold	75.64%	51.17%	77.88%	49.99%	73.91%	49.05%↓	77.61%	49.98%	75.25%	49.39%	64.74%↓	48.46↓	80.74%	51.99%
MMetrics	80.26%	50.92%	79.51%	49.63%	<i>failure</i>	<i>failure</i>	78.43%	49.99%	79.23%	49.45%	79.64%	49.34%	82.67%	52.57%
RFOUT	80.50%	50.88%	80.64%	50.79%	65.09%↓	45.18%↓	78.43%	50.39%	80.02%	50.39%	80.00%	49.38%	83.69%	52.69%
RLR	81.11%	51.23%	81.13%	50.49%	73.20%↓	44.55%↓	78.94%	50.37%	79.51%	50.61%	79.67%	49.25%	83.52%	52.51%
SNOWBALL	80.67%	49.69%	78.77%	49.19%	79.77%	49.54%	80.53%	50.41%	80.05%	50.14%	79.44%	48.40%	82.63%	51.52%
NC+DP	80.29%	50.63%	79.90%	50.50%	69.82%↓	<i>failure</i>	79.05%	49.90%	79.08%	50.00%	79.23%	49.63%	82.71%	53.30%

and ModelReplace are designed for rapid attacks during the later stages when the global model is near convergence, making them less suitable for our setting. Experimental results by Li et al. [15] also indicate that 3DFed should not be launched too early.

Figs. 3 and 4 also display the attack results of FL backdoor methods when faced with defense mechanisms, omitting BAC results for attacks that made the target model unusable. Corresponding MAC results are shown in Table I. The attack

timing limitation of existing FL backdoor attacks becomes more significant when defense mechanisms are deployed. As shown in Fig. 3, on the CIFAR-10 dataset, Neurotoxin fails to achieve ideal BAC across all defense mechanisms, and even the state-of-the-art stealthy FL backdoor attack based on static triggers, 3DFed, only achieves ideal BAC under one defense mechanism. CerP also fails to achieve its ideal attack effect. Fig. 4 illustrates that performance on the CIFAR-100 dataset is

TABLE II
CA AND BAC RESULTS OF DIFFERENT BACKDOOR ATTACKS ON GTSRB

	No Attack CA	ModelReplace CA BAC	Neurotoxin CA BAC	CerP CA BAC	FCBA CA BAC	3DFed CA BAC	ESBA(Ours) CA BAC
FedAvg	88.14%	88.30% 99.64%	89.13% 53.88%	87.10% 73.70%	89.77% 63.97%	89.37% 100%	90.12% 99.73%
Deepsight	88.99%	86.28% 98.67%	88.70% 72.61%	88.89% 66.91%	87.54% 19.17%	87.38% 61.08%	89.90% 92.85%
FLAME	86.60%	85.89% 3.88%	85.28% 52.62%	86.10% 3.49%	85.37% 4.25%	86.22% 3.84%	86.94% 97.04
FLDetector	90.45%	4.96% 0%	89.88% 54.76%	4.80% 0%	85.12% 2.97%	88.02% 9.70%	90.56% 98.70%
Foolsgold	89.84%	88.04% 3.72%	86.22% 21.61%	87.74% 72.27%	88.92% 51.16%	89.14% 7.83%	88.31% 85.09%
MMetrics	88.28%	3.13% 0%	88.82% 24.88%	88.34% 78.08%	86.90% 52.27%	88.67% 7.53%	90.75% 98.93
RFOUT	88.52%	88.42% 100%	86.86% 40.02%	86.01% 64.08%	88.35% 27.01%	86.40% 26.44%	89.15% 93.18%
RLR	89.32%	0.49% 0%	87.75% 47.10%	87.94% 70.46%	87.54% 50.31%	88.58% 51.37%	91.28% 99.87%
SNOWBALL	89.04%	87.46% 3.58%	88.95% 87.36%	89.28% 3.73%	89.81% 4.16%	87.31% 19.15%	90.22% 89.41%
NC+DP	88.69%	85.63% 97.85%	90.00% 33.13%	88.72% 68.03%	88.52% 30.02%	87.05% 34.63%	90.68% 99.83%

TABLE III
CA AND BAC RESULTS OF ESBA ABLATION STUDY

	Deepsight	FLAME	FLDetector	Foolsgold	MMetrics	RFOUT	RLR	SNOWBALL	NC+DP
ESBA-RI	89.33% 92.51%	83.13% 18.73%↓	1.93% 0%↓	86.86% 100%	90.74% 100%	87.75% 100%	87.61% 100%	87.20% 2.90%↓	89.05% 53.14%↓
ESBA-RL	90.08% 85.13%↓	86.45% 99.72%	89.38% 75.68%↓	89.23% 32.76%↓	90.56% 100%	87.69% 32.07%↓	90.18% 79.41%↓	86.01% 12.64%↓	89.24% 63.27%↓
ESBA-RO	89.29% 86.18%↓	85.83% 94.22%	90.11% 90.88%↓	90.47% 65.72%↓	90.81% 99.27%	85.36% 26.38%↓	91.25% 99.36%	87.71% 44.02%↓	89.85% 98.85%
ESBA	89.90% 92.85%	86.94% 97.04%	90.56% 98.70%	88.21% 85.09%	90.75% 98.93%	89.15% 93.18%	91.28% 99.87%	90.22% 89.41%	90.68% 99.83%

slightly better than CIFAR-10 but still insufficient in effectively countering defense mechanisms. CerP, in particular, loses its high BAC due to continuous attacks, reflecting its need for additional constraints on dynamic trigger generation to prevent significant variations that could inhibit backdoor activation. The MAC results in Table I demonstrate the risk of significant performance degradation in the global model when attacks are initiated prematurely. Experimental results indicate that all five advanced FL backdoor attack methods pose a risk of significantly reducing MAC when executed prematurely in a continuous manner.

In summary, current FL backdoor attack methods exhibit limitations regarding the attack timing. In contrast, ESBA can launch and sustain attacks in the early stages of FL without significantly compromising the MAC of the global model while achieving an optimal BAC for the implanted backdoor. The MAC results in Table I demonstrate that ESBA, whether facing FedAvg or other defense mechanisms, does not noticeably impact the global model's MAC. Figs. 3, and 4 further illustrate that ESBA can successfully implant a backdoor with an ideal BAC in the global model, regardless of facing FedAvg or other defense mechanisms.

Our experiments on the GTSRB dataset illustrate a scenario where an attacker needs to perform attacks as early as possible and succeed before the global model approaches convergence if the model trainer halts training once convergence is near. In our FL scenario, the model trainer can terminate training when the global model's loss falls below 0.5, and the decrease in loss between rounds is less than 0.01, indicating its near convergence. Under such conditions, existing FL backdoor attack methods often fail to achieve their desired effect. The results are presented in Table II, where we bolded successful attack results with ideal BAC. Only ModelReplace achieves an ideal BAC under certain defenses. However, due to its blind amplification of backdoor model updates, it risks significantly reducing the global model's MAC. FCBA, Neurotoxin and CerP fail to implant the backdoor in time, even when facing the FedAvg method, which lacks resistance against backdoor attacks. Notably, ESBA effectively bypasses any defense mechanism, implanting a backdoor with

ideal BAC in the global model without compromising MAC before the global model near convergence. In summary, ESBA demonstrates excellent stealth and remarkable performance in attack timing.

C. Ablation Study of ESBA

We conducted ablation experiments on the GTSRB dataset using three variants of ESBA, each of which removes a specific mechanism. The experimental results are shown in Table III, where significantly reduced BAC values compared to ESBA are highlighted with arrows. Similar to the comparative experiments, we assume that the server ends training of the global model when its loss is less than 0.5, and the decrease in loss compared to the previous round is less than 0.01. ESBA-RI denotes ESBA without the inertia poisoning mechanism. The results show that The inertia poisoning mechanism helps enhance ESBA's ability to bypass defense mechanisms and reduces its negative impact on the global model's CA. When the backdoor model updates are no longer aligned through the inertia poisoning mechanism, ESBA fails to bypass defenses such as FLAME, FLDetector, and SNOWBALL, and may also lead to a decrease in the global model's CA. ESBA-RL and ESBA-RO denote ESBA variants without constraint loss and centralized optimization mechanisms for backdoor model updates, respectively. The purpose of these two mechanisms is to make backdoor model updates more difficult for defense mechanisms to detect. The experimental results demonstrate that removing either of these mechanisms decreases ESBA's overall BAC attack performance against potential defense mechanisms. In conclusion, ESBA can execute stealthy FL backdoor attacks before the global model has nearly converged due to the interplay of these three mechanisms. Removing any of these three mechanisms would significantly reduce ESBA's ability.

D. Impact of Hyper-Parameters

This section designs experiments to examine the impact of different parameter settings on ESBA's early attack capability.

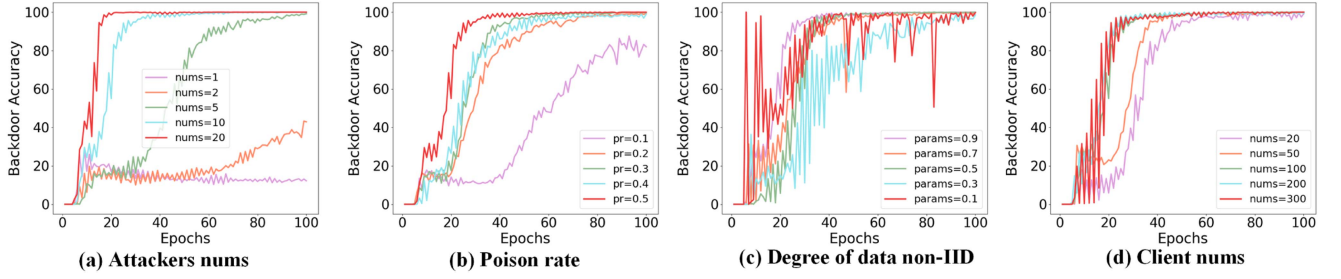


Fig. 5. BAC results of the ESBA attack under different settings on CIFAR-10 (%).

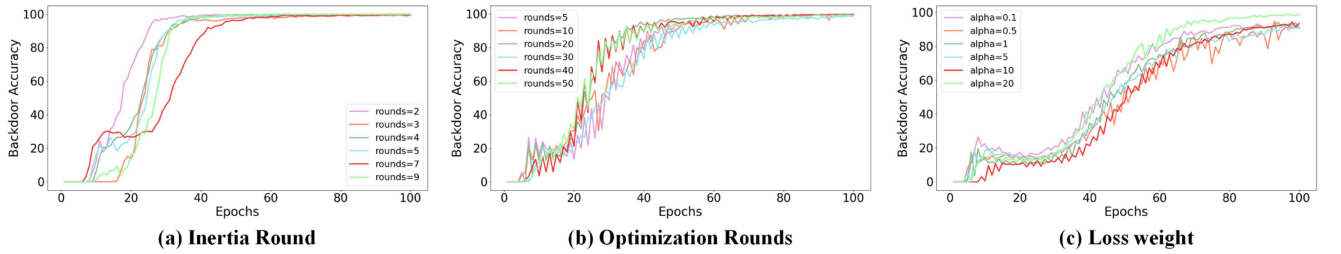


Fig. 6. The impact of different parameter settings on ESBA's BAC performance on CIFAR-10. (%)

Fig. 5 presents the attack results of ESBA within 100 rounds under varying numbers of attackers, poisoning rates, degrees of data non-IID, and numbers of clients. By default, the number of attackers is set to 10% of the total number of clients, the poisoning rate is set to 0.5, the degree of non-IID data is controlled using a Dirichlet distribution with a parameter of 0.9, and the total number of clients is 100. Experimental results indicate that an insufficient number of malicious clients or a low poisoning rate may prevent ESBA from achieving the desired attack effect. ESBA maintains robustness under different levels of data heterogeneity and varying numbers of clients, though excessively high data heterogeneity can negatively impact the stability of backdoor implantation.

Fig. 6 illustrates the impact of different ESBA's parameter settings on the effectiveness of ESBA attacks. The default settings for inertia accumulation rounds, centralized optimization rounds for backdoor model updates, and constraint loss weights are 5, 1, and 1, respectively. The experimental results shown in Fig. 6(a) demonstrate that the inertia accumulation rounds should not be set too large. The experimental results show that the constraint round with the least historical information achieved the best performance. Combined with the ablation results, BAC increases when the inertia constraint is removed. These indicate that the inertia poisoning mechanism imposes additional constraints on the parameter updates of the backdoor model, thereby limiting its performance on the backdoor task to some extent. The primary purpose of centrally optimizing backdoor model updates is to address the potential adverse effects caused by the constraint loss. The experimental results indicate that only a small amount of optimization epochs is required to handle this issue effectively. Regarding the experimental results for constraint loss weights, it is shown that smaller weight settings are more conducive

to ESBA achieving optimal attack effectiveness. In contrast, larger weights may potentially impact the effectiveness of ESBA attacks. This is because constraint loss constrains modifications to dimensions that are more important for the backdoor task.

E. Attacking Large Language Models

We tested whether ESBA can still maintain its strong attack performance when the FL framework is used for large language model fine-tuning tasks. This scenario differs significantly from traditional small-model tasks. Due to the rapid fine-tuning speed of large models, the global model approaches convergence in few rounds. Table IV presents detailed attack results, with total training rounds for IMDB and Obscene being 5 and 10, respectively. We use arrows to highlight the disappointing MAC and BAC results. Due to the global model's rapid convergence in this scenario, various comparative methods can fully exploit their attack effects. Only the ModelReplace method, due to its blind scaling mechanism, still has the potential to harm the target model's MAC significantly. The other comparative methods no longer harm the target model's MAC. Although 3DFed, focused on stealthy, can handle many defense mechanisms, it is less effective in terms of stealth compared to Neurotoxin. This reflects that due to the unique characteristics of the large language model fine-tuning scenario, some mechanisms designed for stealthy attacking traditional small models may not be suitable. Notably, ESBA is also capable of handling all defense mechanisms and achieving success in this scenario. Our experiments in the large language model fine-tuning scenario have thoroughly validated ESBA's excellent versatility. In conclusion, regardless of whether the global model is nearing convergence or not, or whether the target model is a traditional small model or a large

TABLE IV
CA AND BAC OF DIFFERENT BACKDOOR ATTACKS ON FINE-TUNING TASKS FOR LARGE LANGUAGE MODEL

Dataset	Defense	No Attack CA	ModelReplace CA	BAC	Neurotoxin CA	BAC	3DFed CA	BAC	ESBA(Ours) CA	BAC
IMDB	FedAvg	91.78%	83.53%↓	100%	91.64%	99.92%	91.28%	99.97%	91.54%	100%
	Deepsight	91.58%	91.24%	99.44%	91.38%	99.96%	90.96%	99.68%	91.29%	96.63%
	FLAME	91.22%	91.16%	6.41%↓	91.21%	98.39%	91.24%	4.60%↓	91.14%	94.89%
	FLDetector	91.72%	88.00	98.96%	91.70%	99.47%	89.94	99.97%	91.94%	100%
	Foolsgold	91.54%	91.31%	3.91%↓	91.58%	99.92%	91.37%	5.33%↓	91.67%	99.97%
	MMetrics	91.49%	51.64%↓	38.64%↓	91.40%	99.93%	91.19%	54.96%↓	91.39%	99.19%
	RFOUT	91.70%	87.21%↓	100%	91.48%	99.46%	90.99%	100%	92.09%	99.99%
	RLR	91.72%	89.44%	100%	91.42%	99.70%	90.06%	99.99%	91.96%	100%
	SNOWBALL	91.58%	90.37%	99.97%	91.35%	91.68%	91.46%	0.41%↓	91.37%	100%
	NC+DP	91.70%	91.29%	98.74%	91.67%	99.80%	90.69%	99.99%	91.86%	100%
Obscene	FedAvg	95.02%	95.21%	97.76%	94.93%	97.60%	94.47%	97.42%	95.30%	97.60%
	Deepsight	95.11%	95.30%	96.86%	95.11%	97.42%	95.02%	97.42%	95.21%	97.42%
	FLAME	95.21%	94.75%	11.78%↓	94.75%	13.07%↓	95.02%	11.60%↓	94.84%	94.65%
	FLDetector	95.02%	93.37%	98.15%	95.21%	97.60%	95.67%	97.23%	95.48%	97.60%
	Foolsgold	95.39%	94.93%	10.86%↓	94.75%	97.60%	94.65%	11.04%↓	95.39%	97.60%
	MMetrics	94.47%	50.09%↓	64.08%↓	95.11%	97.42%	94.65%	69.24%↓	95.30%	97.60%
	RFOUT	94.84%	88.48%↓	97.60%	94.56%	97.60%	95.02%	97.60%	94.93%	97.60%
	RLR	94.93%	95.58%	97.79%	95.02%	97.60%	95.21%	97.60%	95.02%	97.60%
	SNOWBALL	95.11%	94.56%	97.79%	94.56%	96.86%	94.75%	22.89%	95.21%	97.6%
	NC+DP	94.84%	95.02%	97.42%	95.21%	97.42%	94.01%	94.01%	95.11%	97.42%

TABLE V
CA AND BAC RESULTS OF ATTACKING PDFM

Dataset	No Attack CA	ModelReplace CA	BAC	Neurotoxin CA	BAC	CerP CA	BAC	FCBA CA	BAC	3DFed CA	BAC	ESBA CA	BAC
GTSRB	91.67%	90.03%	3.60%	90.78%	3.67%	91.99%	4.67%	91.40%	4.04%	91.19%	3.70%	90.59%	3.23%
CIFAR-10	79.98%	79.32%	10.43%	79.26%	9.71%	78.58%	11.49%	79.20%	22.42%	79.13%	10.14%	79.35%	10.22%
CIFAR-100	50.38%	49.22%	1.01%	49.62%	3.94%	49.73%	2.01%	49.96%	0.92%	48.94%	0.91%	50.36%	1.84%
IMDB	91.07%	91.26%	5.99%	91.18%	5.00%	Inapplicable	Inapplicable	Inapplicable	Inapplicable	91.28%	4.12%	91.52%	4.92%
Obscene	94.48%	94.84%	11.41%	94.56%	10.68%	Inapplicable	Inapplicable	Inapplicable	Inapplicable	95.12%	14.91%	94.56%	11.23%

language model, ESBA can successfully bypass defenses and implant a backdoor into the target global model.

Our experiments also show that most defense mechanisms designed for traditional small-model tasks fail in this scenario, revealing severe backdoor threats in large language model fine-tuning tasks. Researchers focusing on FL defense mechanisms urgently need to develop effective defenses tailored to this scenario.

F. Effect of PDFM

Table V presents the MAC and BAC results of using four advanced FL attack methods and ESBA to attack PDFM. The five datasets shown from top to bottom in the table were trained for 40, 220, 220, 5, and 10 rounds, respectively, enabling complete convergence of the global model within these rounds. From the above experiments on the effectiveness of ESBA and four representative FL backdoor attack methods, it is evident that existing FL backdoor defense mechanisms are unable to counter ESBA and are susceptible to circumvention by current FL backdoor attack methods or jeopardizing the global model's ability to learn the original task. Observing the experimental results in Table V, PDFM effectively protects the global model from attacks by ESBA and four representative FL backdoor attack methods. The maximum BAC achieved across these attacks does not exceed 15%, indicating that PDFM successfully filters out backdoor model updates with anomalous statistical distributions, rendering the attacks ineffective. Additionally, the MAC

results for the global model under attack show no significant decline compared to those not under backdoor attack, indicating that PDFM effectively protects the global model's learning of the original task from being compromised by backdoor attacks.

For weighted or filter-then-mean aggregation, the aggregation process can be written as $G_t = G_{t-1} + \eta \cdot \sum_{i=1}^N \gamma_i \Delta W_{t,i} / \sum_{i=1}^N \gamma_i$. The constraint loss in (7) and the centralized optimization in (11) help control the magnitude of $\Delta W_{t,adv}$, while the inertia mechanism aligns its direction with the global update trend. Therefore, ESBA can reduce both direction-level and norm-level abnormality. For clipped mean aggregation, each update is first clipped as $\widehat{\Delta W}_{t,i} = \Delta W_{t,i} \cdot \min(1, C/|\Delta W_{t,i}|^2)$, and the server aggregates the clipped updates. Therefore, if a malicious update has a bounded norm, it can participate in aggregation without being clipped. Even when clipping is triggered, the update is only multiplied by a scalar factor $C/|\Delta W_{t,adv}|^2$, and its inertia-aligned direction is preserved. In ESBA, the constraint loss in (7) suppresses excessive parameter changes, while the centralized optimization in (11) constrains the aggregate perturbation of malicious updates. However, inertia alignment mainly constrains the direction of malicious updates, but it does not ensure that their parameter distributions match those of benign updates. Due to the difference between the backdoor objective and the benign task objective, malicious updates accumulate different mean and variance patterns in inertia-related dimensions. PDFM exploits this limitation by clustering $d_i = (\text{mean}(H_i^*), \text{var}(H_i^*))$ to identify statistically anomalous updates.

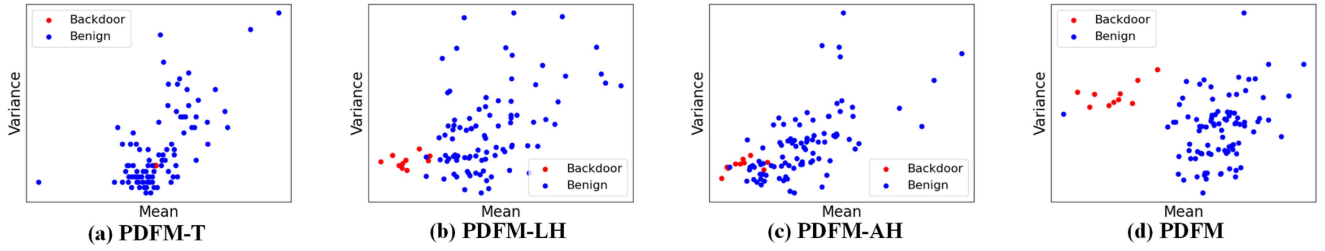


Fig. 7. Visualization of model updates statistical distribution.

TABLE VI
RESULTS OF PDFM ABLATION STUDY

Method	Dataset	Neurotoxin		3DFed	
		CA	BAC	CA	BAC
PDFM-T	GTSRB	92.10%	3.12%	91.01%	92.23%↑
	CIFAR-10	79.69%	36.78%↑	79.93%	66.45%↑
	CIFAR100	50.22%	1.42%	49.18%	91.95%↑
PDFM-LH	GTSRB	90.86%	83.37%↑	90.64%	70.04%↑
	CIFAR-10	80.15%	9.69%	78.62%	65.77%↑
	CIFAR100	49.38%	1.07%	49.42%	0.87%
PDFM-AH	GTSRB	92.28%	98.90%↑	91.84%	98.11%↑
	CIFAR-10	79.12%	21.06%	79.13%	10.37%
	CIFAR100	50.13%	45.77%↑	49.36%	1.24%

G. Ablation Study of PDFM

We used three variants of PDFM to test the significance of each component in PDFM's design. PDFM-T is a variant that only considers the current epoch's model updates to test the necessity of aggregating historical updates. PDFM-LH is a variant that uses only the previous round's global model update direction instead of the accumulated inertia direction to filter dimensions, which tests the need for aggregating multiple epochs' global model update directions. PDFM-AH is a variant that filters using the mean and variance of all dimensions without selecting specific dimensions based on inertia directions. It is aimed at testing the necessity of selecting specific dimensions. Table VI displays the results of each variant against Neurotoxin and 3DFed in image recognition tasks, where the BAC results that are significantly larger compared to PDFM are highlighted with arrows. The results indicate that any one of these variants weakens the defense effectiveness against Neurotoxin and 3DFed, proving the importance of each tested design component. Fig. 7 provides visualizations of the parameter distribution computed by PDFM and its three variants before clustering, with the horizontal axis as the mean and the vertical axis as the variance. Each point represents the parameter distribution of a model update, where points for backdoor model updates are marked in red and benign model updates in blue. Observing the visualized parameter distributions reveals that only PDFM's calculated results are distinguishable by clustering algorithms. In summary, PDFM's design of considering client historical updates and selecting dimensions based on accumulated inertia direction is meaningful.

TABLE VII
PERFORMANCE OF ESBA AND PDFM UNDER DIFFERENT FL SETTINGS

Dataset	FL setting	No Attack CA	Trend		PDFM	
			CA	BAC	CA	BAC
GTSRB	asynchronous	88.40%	89.56%	99.98%	88.95%	4.57%
	synchronous	88.14%	90.12%	99.73%	90.59%	3.21%
CIFAR-10	asynchronous	80.80%	81.17%	99.99%	78.52%	10.19%
	synchronous	81.1%	83.45%	100%	79.35%	10.22%
CIFAR-100	asynchronous	50.96%	51.22%	100%	49.80%	0.98%
	synchronous	50.70%	52.18%	100%	50.36%	1.84%

H. Evaluation Under Other Practical Attack Scenarios

To further evaluate the practicality of our attack and defense methods, we conduct experiments under an asynchronous FL setting. In this setting, we assign a delay to each uploaded update: 70% of updates have no delay, 20% are delayed by one round, and 10% are delayed by two rounds. Therefore, the aggregated updates may be generated from different historical global models, which simulates stale updates caused by heterogeneous client response times. To reduce the influence of stale updates, we further apply a staleness-aware weight $\alpha(s_{delay}) = 1/(1 + s_{delay})$ during aggregation, where s_{delay} is the number of delayed rounds. Table VII reports the performance of ESBA and PDFM under synchronous and asynchronous FL settings. In the asynchronous setting, ESBA still achieves nearly 100% BAC on all three datasets. Meanwhile, their CA results are higher than the no-attack baseline. This indicates that asynchronous client delays and stale updates do not weaken the attack effectiveness of ESBA. PDFM also remains effective under asynchronous FL. When PDFM is deployed, the BAC is reduced to 4.57%, 10.19%, and 0.98% on GTSRB, CIFAR-10, and CIFAR-100, respectively. At the same time, PDFM maintains acceptable CA. Compared with the synchronous setting, the asynchronous setting mainly affects CA, while having little impact on BAC. This suggests that stale updates introduce additional noise and slightly affect the main-task performance. For ESBA, the CA decreases under asynchronous FL. However, the attack BAC is almost unaffected and remains close to 100% on all datasets. For PDFM, the CA also decreases under asynchronous FL, but the defense effectiveness remains stable. These results indicate that asynchronous FL mainly causes a slight CA degradation, but it does not weaken the backdoor implantation capability of ESBA or the backdoor suppression capability of PDFM.

We also evaluate the performance of ESBA and PDFM under another practical FL scenario using the Tiny-ImageNet dataset.

TABLE VIII
CA AND BAC RESULTS (%) OF DIFFERENT BACKDOOR ATTACKS ON TINY-IMAGENET

	No Attack CA	ModelReplace CA BAC	Neurotoxin CA BAC	3DFed CA BAC	CerP CA BAC	FCBA CA BAC	ESBA(Ours) CA BAC	ESBA-C CA BAC	ESBA-EC CA BAC
FedAvg	39.08	25.28 100	38.49 98.53	38.03 99.95	37.22 99.23	38.27 57.89	38.98 98.29	39.03 96.40	38.25 98.16
DeepSight	38.98	34.13 99.95	38.80 95.25	0.50 0	38.46 99.72	37.09 62.25	38.57 92.94	39.20 95.89	39.02 97.86
FLAME	37.27	36.25 67.96	37.03 17.59	36.85 65.04	35.14 17.98	35.15 20.62	37.47 97.06	36.27 81.26	37.45 94.77
FLDetector	38.55	31.11 99.99	38.16 64.12	36.47 77.61	39.03 99.52	37.50 22.03	38.62 83.40	39.62 49.07	38.25 50.28
Foolsgold	39.71	34.17 99.98	38.84 87.62	36.71 97.20	37.41 98.11	37.58 49.47	38.73 91.44	39.02 81.50	38.38 86.20
MMetrics	37.08	0.50 0	37.12 99.45	36.53 98.53	36.98 99.57	36.98 71.61	38.16 99.56	37.85 96.81	38.14 99.10
RFOUT	39.72	26.70 100	38.19 98.51	34.18 100	38.29 98.48	38.15 73.35	39.29 97.49	39.46 93.47	39.16 94.41
RLR	39.11	24.93 100	38.94 99.34	35.64 99.60	37.87 99.69	37.83 63.57	39.27 95.28	38.36 93.72	38.28 96.62
SNOWBALL	37.87	37.23 85.10	38.07 97.92	34.50 99.81	37.07 99.49	36.06 26.72	37.15 90.11	38.14 64.05	37.38 93.12
NC+DP	39.44	25.83 100	39.10 97.75	0.50 0	38.44 99.77	38.64 67.42	39.42 97.64	39.90 94.79	39.60 94.41
PDFM(Ours)	38.56	37.84 0.83	36.98 0.97	38.10 0.44	37.73 0.59	38.75 0.91	37.17 0.83	36.83 0.44	37.18 0.79

In this scenario, clients may be offline, and the server randomly selects a subset of received model updates for aggregation. To simulate this setting, we assume that in each round, both benign and malicious clients have a 0.2 probability of failing to upload their model updates. The server then randomly selects 10 of the received updates for aggregation. The FL training lasts for a total of 300 rounds. Additionally, we explore the potential of combining ESBA with dynamic triggers and edge-case FL backdoor attacks. The experimental results are presented in Table VIII. In particular, ESBA-C refers to an ESBA variant that adopts the same dynamic trigger as CerP, and ESBA-EC denotes an ESBA variant that uses the edge-case strategy proposed by Wang et al. [17]. For PDFM, the results show that it remains effective in this scenario, consistently achieving low BAC and high CA under various backdoor attacks. As for ESBA, the results demonstrate that it can successfully implant backdoors into the global model under all defense strategies except PDFM. Compared to other attack methods, ESBA shows advantages in both preserving global model CA and bypassing defense mechanisms. The results of ESBA-C and ESBA-EC indicate that ESBA is extensible. Compared to CerP, ESBA-C achieves better preservation of the global model's CA and successfully bypasses FLAME. However, ESBA-C and ESBA-EC exhibit noticeable drops in BAC when attacking specific defense mechanisms, suggesting that further targeted optimization is needed when integrating ESBA with other advanced strategies. We will conduct an in-depth study of ESBA's extensibility in future work.

VII. CONCLUSION

This paper investigates the problem of backdoor attacks in short-term FL scenarios and explores a new parameter distribution-based anomaly detection metric for identifying backdoor model updates. We first propose a novel FL backdoor attack method, ESBA. ESBA can persistently launches attacks throughout the entire FL training process and successfully injects backdoors before the global model approaches convergence by leveraging a customized loss function, an inertia poisoning mechanism, and centralized optimization of backdoored model updates. In addition, we propose a novel FL defense mechanism, PDFM, which utilizes a novel anomaly detection metric and the HDBSCAN clustering algorithm to detect backdoor model updates. Extensive experiments demonstrate the effectiveness and superiority of both ESBA and PDFM. For future work, we observe the potential of combining ESBA with more advanced

FL backdoor attack strategies. We plan to develop more powerful FL backdoor attacks along with their corresponding defense mechanisms.

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